

NSBV BC2001: Laboratory in Neuroscience
Barnard College, Spring 2026

Students enroll in one lab section. All labs meet in Diana LL104 unless otherwise announced.

Section	Day	Time	Instructor	Teaching Assistant
1	Thursday AM	9:10 a.m. – 1:00 p.m.	Prof. Shuk Tsoi	Dana (Caroline) Villamin
2	Thursday PM	1:40 – 5:30 p.m.	Prof. Luca Iemi	Ritika Rohatgi
3	Friday AM	9:40 a.m. – 1:30 p.m.	Prof. Luca Iemi	Jessica Reschny

Contact Information for Course Instructors, Teaching Assistants, and Computing Fellows

Name	Email	Student Hours
Luca Iemi	llemi@barnard.edu	See Canvas/Pages/Student Hours Student hours will begin on the week of 2/2.
Shuk Tsoi	stsoi@barnard.edu	
Dana (Caroline) Villamin	dcv2109@barnard.edu	
Ritika Rohatgi	rr3578@barnard.edu	
Jessica Reschny	jnr@2152@barnard.edu	

Course Communication: For general questions regarding course logistics, assignment instructions and due dates, or course content, please use Canvas/Discussions. For personal matters regarding grades, attendance, accommodations, or emergencies, please contact your instructor via email.

Prerequisite: Successful completion of an introductory neuroscience lecture course.

General Education Requirements: This lab class, in conjunction with a completed introductory neuroscience lecture class, fulfills the Barnard College Foundations for one semester of a science with laboratory requirement. This lab class also fulfills the Technologically and Digitally Mode of Thinking.

Major Requirement: This lab is a neuroscience class required of all Barnard Neuroscience and Behavior majors. This class may also be used as a lab course option for Barnard Psychology majors.

Course Materials: There is no required material you must purchase for this lab. All readings and protocols will be provided to students electronically and in paper form in the lab. **You should bring your laptop computer to the lab each week in order to access electronic resources. On Canvas, all materials and assignments will be sequentially ordered under the “Modules” tab.**

Course Description and Learning Outcomes: This class will introduce students to a sampling of experimental methods commonly used in neuroscience research. Neuroscience is an interdisciplinary field that draws from the foundational sciences, including biology, chemistry, physics, psychology, computer science, and math. Techniques used in neuroscience research can vary broadly and may encompass molecular and cellular biochemistry, anatomical investigations, animal behavior, human cognition, electrophysiology, computational modeling, and other fields. Upon successful completion of this course, students should be able to:

1. Understand both structural and functional mammalian neuroanatomy
 - Gross anatomy of sheep brains
 - Microscopic anatomy of rodent brains
 - Cortical anatomy of human brains
2. Formulate a testable hypothesis based on published studies and preliminary data
 - Search scholarly databases for relevant scientific literature
 - Read primary research papers and review articles
 - Cite papers using a reference manager program, such as Zotero or Endnote
3. Conduct a properly controlled experiment following safety and ethical guidelines
 - Complete lab safety training
 - Complete human subjects protection training
 - Adhere to all safety and ethical guidelines
4. Collect data and conduct analysis using statistical methods
 - Conduct immunohistochemical staining of rodent brains and microscopy analysis
 - Conduct and analyze invertebrate electrophysiological recordings
 - Conduct and analyze human EEG recordings
 - Implement relevant experimental design and statistical analyses
 - Use MATLAB to analyze data and report results
5. Interpret data within the context of the neuroscience literature
 - a. Understand results as they relate to experimental design and published studies
 - b. Propose future studies that build on obtained results
6. Communicate scientific information through lab assignments
 - Write a lab report that includes sections of a scientific manuscript: abstract, introduction, methods, results, discussion, and references

Grading:

- Entry Ticket Assignments = 15%
- Exit Ticket Assignments = 20%
- Homework Assignments = 20%
- Lab Report
 - Draft = 2%
 - Final version = 13%
- Neuroanatomy Exam = 15%
- Lab Final Exam = 15%

Entry tickets are assignments that must be completed prior to lab each week to help students prepare for the lab activities. Details of these assignments will be posted on Canvas.

Exit tickets are assignments related to completion of in-lab work, submission of data collected, and lab cleanup. Descriptions of exit tickets will be provided during the lab and posted on Canvas. Exit tickets are due at the end of each lab.

Homework assignment descriptions will be made available in the lab and posted on Canvas. Some assignments may be started in class, while others will be completed entirely outside of class. Homework assignments are due by the date and time indicated on Canvas.

Lab Report Students will write one full lab report in the form of a published research article. This report will be based on the electrophysiology project conducted in class. Students will submit a draft report and feedback from their instructor prior to submitting the final draft.

MATLAB digital notebook. You are required to keep a digital notebook related to your MATLAB training and assignments. Notes might include details regarding how to define variables, create vectors and matrices, or use new functions. You must bring this notebook with you to every lab class. The instructors and TAs will occasionally check your notebook to ensure that you are staying on track with your MATLAB learning. You will be allowed to use this notebook during the MATLAB portion of the final exam.

MATLAB folder. On your computer, you must also create a folder where you will store files related to your MATLAB assignments. If you install the desktop version of MATLAB on your computer, a “MATLAB” folder will usually be created for you in either your “Home” folder or your “Documents” folder.

On your computer (or Google Drive, or MatLab Drive), you must also create a folder where you will store files related to your MatLab assignments. If you install the desktop version of MatLab on your computer, a “MatLab” folder will usually be created for you in either your “Home” folder or your “Documents” folder.

Schedule. This schedule is subject to change. It is your responsibility to attend the lab and effectively communicate with your instructors, TAs, and peers so that you will be aware of any changes in the course.

Date	Pre-Lab Entry Ticket	Lab	Assignment
Week 1 1/20-1/23		Lab Introduction <i>Introductions, Course Overview, Lab Safety</i>	
Week 2 1/26-1/30	TC2150 Safety Training and Certification & Course Policy Contract	Neuroanatomy 1 <i>Modules 1 -8</i>	Exit Ticket: Worksheet HW: Neuroglossary 1
Week 3 2/2-2/6		Neuroanatomy 2 <i>Modules 9 - 13</i>	Exit Ticket: Worksheet HW: Neuroglossary 2
Week 4 2/9-2/13	Matlab Tutorial Experimental Design & Statistics	Lab Skills 1 <i>MATLAB, experimental design, and statistics workshop</i>	Exit Ticket: Matlab workshop HW: Study for Exam!
Week 5 2/16-2/20		Neuroanatomy Exam	
Week 6 2/23-2/27	Intro to Pipettes & Buffers	Lab Skills 2 <i>Pipettes, solutions, and buffers</i>	Exit Ticket: lab activities HW: Buffer Titration
Week 7 3/2-3/6	Histology	Histology 1 <i>Primary antibodies, practice mounting sections</i>	Exit Ticket: IHC
Week 8 3/9-3/13	Preparing Chemical Solutions	Histology 2 <i>DAPI stain, mount and coverslip slides</i> Lab Skills 3 <i>Brightfield Microscopy</i>	Exit Ticket: IHC & mouse atlas HW: Histology Unit Review (due after spring break)
Week 9 3/16-3/20		Spring Break	Relax & Have Fun!!
Week 10 3/23-3/27	Download Java & FIJI	Histology 3 <i>brain atlas workshop, FIJI image analysis</i>	Exit Ticket: Fluorescent Microscopy Histology Data Analysis

Date	Pre-Lab Entry Ticket	Lab	Assignment
Week 11 3/30-4/3		EEG1 <i>Intro to EEG and test run</i>	Exit Tickets: EEG biophysics; EEG Artifacts
Week 12 4/6-4/10	Fonseca et al paper & quiz	EEG2 <i>Experiment</i>	5 Exit Tickets HW: Single-subject analysis & Questions
Week 13 4/13-4/17		EEG3 Single subject Analysis	Exit Tickets: Group-level data analysis
Week 14 4/20-4/24	Action Potential Propagation	Electrophysiology 1 <i>Earthworm recordings</i>	HW: Analysis Lab Report Draft due 4/24
Week 15 4/27-5/1	Worm study proposal	Electrophysiology 2 <i>Earthworm recordings</i>	Proposal presentation
Week 16 5/8-5/14		Final Exam <i>Date and location TBD</i>	Lab Report Final Draft Due (5/5)

Class Policies

Lab Safety: The neuroanatomy, histology, and electrophysiology modules of this lab course are wet lab units that involve working with tissues and chemicals. All students must complete the TC2150 Barnard Laboratory Safety Training within the first lab session.

It is imperative that you come to lab each week prepared for safety. This includes adhering to the following rules:

1. Wear closed toe shoes.
2. Wear clothes that cover your legs.
3. If you have long hair, bring hair ties/clips so you can keep your hair tied back.
4. Eating and drinking are never permitted inside the lab classroom. You may keep a water bottle and snacks in the hallway outside the classroom and take brief breaks to drink and eat.
5. You will be provided with a lab coat and safety goggles to use and store in the lab.
6. You will be provided with gloves to use in the lab.

Attendance and Participation: Because successful completion of the lab requires active and collaborative participation, students are expected to attend their registered lab section each week.

Laboratory attendance is mandatory. Labs are a key place in your educational experience where you can engage in hands-on experiential learning. Learning in these classes happens by doing structured activities in the lab. For these reasons, your attendance is necessary to meet the learning objectives of the course.

Excused absence due to a medical/family emergency: If an emergency arises that prohibits you from attending lab (e.g., serious illness), it is possible that you may be excused from lab and scheduled to attend a make-up lab only *if* you follow the series of steps outlined below.

2. Email your lab instructor *before* your scheduled lab section meets. If the nature of your emergency is extenuating (e.g., hospitalization) and prevents you from contacting your instructor before class, please email your instructor within 24 hours of your emergency.
3. You will receive a reply to your email from your lab instructor by the end of the school day. You must respond to this email by the following morning to confirm your make-up lab option. Make-up labs are subject to the availability of an instructor, classroom space, and protocol time constraints. You may need to complete lab activities that are different from those conducted during lab class.
4. Any assignment due on the day of your absence must still be turned in prior to the start of your regularly scheduled lab unless the nature of your emergency prevents you from doing so.
5. No more than TWO excused emergency absences and make-ups will be permitted no matter the circumstances. In the case of an extended absence, we will work together with your Class Dean to develop a plan to either drop the course, withdraw, or take an incomplete so that you can complete the course when your health permits.

Excused absence due to religious observation: If observance of a major religious holiday prevents you from attending lab, it is your responsibility to attend a make-up lab during the same week by scheduling arrangements with your lab instructor in advance. *You must notify your lab instructor about a religious observation conflict by Friday, February 13th.* If you need to reschedule more than one lab over the course of the semester, you are encouraged to register for a different lab section during the course shopping period.

Unexcused absence: Failure to adhere to the above policies for excused absences will result in an unexcused absence. Labs will not be rescheduled for unexcused absences, and assignments associated with the missed lab will not be graded. If you have one unexcused absence from a lab, your final course grade will be reduced by a full letter grade (for example: "A" to "B"). Two or more unexcused absences will result in a failing course grade.

Tardiness: Attendance is recorded every week. If you are late to class, this will be noted. For more than two late arrivals, your semester grade will be lowered by 1% per occurrence.

Participation: Participation is assessed by your weekly entry and exit ticket assignments. It includes coming to class on time, being prepared for lab by completing homework and pre-lab exercises, wearing appropriate lab attire, actively contributing to group work and discussions, making thoughtful and respectful comments during class, cooperating with group members, and cleaning up after lab is finished.

Group work: Group work is a core component of science, whether it is in the undergraduate classroom or in a research lab. You will work in small groups throughout the semester. The participants of each group are required to contribute equally to all group activities and assignments and to treat one another with respect.

Assignment Policies:

- Please bring your laptop computer and a charger to lab each week. There are in-class assignments that require a computer.
- Assignment instructions will be available on Canvas, and assignment submissions must be completed/uploaded on Canvas.
- Assignments will be graded anonymously. Do not type your name anywhere on your assignment.
- *It is your responsibility to view your submitted assignment online to verify that it uploads correctly and is in an accessible format (i.e., PDF, Word, Powerpoint, MATLAB, etc., as indicated in the assignment instruction). If your upload is missing or you upload a file that can't be opened by the instructor or you upload the wrong file, you will receive a zero.*
- **Late entry tickets will immediately incur a deduction penalty of 50% per day. Other assignments will have a penalty of 10% per day. You are allowed up to a total penalty of 50%.**
- *Assignments late by more than one week will not be accepted and will receive a zero.*
- There is no guarantee that late assignments will be graded in a timely manner or handed back prior to the final exam. If you want feedback on your assignments, you must submit your assignments on time.
- **There will be no extra credit assignments.**

Exam Policies:

- A student who is late on an exam day may not have extra time to complete the exam.
- Exams will not be handed out to any student who arrives at the lab after another student has completed and handed in their exam.
- If you are absent for an exam, you must provide appropriate documentation for your absence. A make-up exam must be taken within one week of the original test date. Make-up exams may be different in format from the exams administered in the class. Make-up exams will not be offered for unexcused absences.
- **The neuroanatomy quizzes include a lab practical and must be taken in the lab during class time.**
- The lab final exam will be scheduled by the Registrar to take place during the final exam period (May 8-14).

Grade Policies:

- Individual assignments and exams are not curved. If necessary, the instructors will make a decision on curving final course grades at the end of the semester. Historically, grades for this lab course have not been curved because most students perform well as long as they attend class consistently, engage mindfully in the work, complete the assignments on time, and study adequately for the exams.
- There is no rounding up of final grades. Please do not email your instructor to ask about rounding up your grade.
- The grading scale is as follows:
 - 93.0-100 (A)
 - 90.0-92.9 (A-)
 - 87.0-89.9 (B+)
 - 83.0-86.9 (B)
 - 80.0-82.9 (B-)
 - 77.0-79.9 (C+)
 - 73.0-76.9 (C)
 - 70.0-72.9 (C-)
 - 60.0-69.9 (D)
 - 0-59.9 (F)

Ethical scientific conduct: Unethical scientific conduct (such as fabrication of data, plagiarism, etc.) will result in automatic failure in this class and notification to your Class Dean and the Honor Board.

You will be assessed on how you carry out the scientific method and how you interpret data. You will never be assessed on whether you obtain expected results. Do not make up data. It is never appropriate.

Writing assignments will be scanned by Turnitin plagiarism software. Assignments that are not uploaded on Canvas can't be scanned and will thus not be graded and will receive a zero.

Some assignments ask you to self-certify that you have completed a task. If you certify a task before it is completed, you will lose points on the assignment.

General lab policies: The general college policy is that one academic credit demands 3 hours of work per week. This lab is worth 3 credits; this translates into 9 hours of work per week. The 4 hours of in-class lab time are included in these 9 hours. The remaining time is devoted to time spent studying and completing homework assignments outside of class.

Students are strongly encouraged to utilize weekly student hours to discuss anything course-related with their lab instructors and teaching assistants. Student hour details (days/times/locations) for each instructor and TA are listed on the course syllabus. Appointments are not necessary for student hours. Just drop in!

If you wish to discuss a personal matter with your instructor, you should schedule a one-on-one appointment.

If you have general questions about the course or the assignments, you can post your questions in the Canvas/Discussions. Do not email your instructor or TA with general lab questions. We will not answer general questions by email because it only serves to benefit the student who sent the email. If you post the question in the discussion, we will answer your question there. This way, the entire class can learn from everyone's inquiries.

You are responsible for verifying that emails you send have indeed been sent and that assignments you submit have indeed been uploaded in the appropriate format. These are simple technical confirmations and excuses made otherwise will not be accepted.

CARDS accommodations must be requested through the AIM portal. Accommodations are considered active once they are acknowledged on AIM. Accommodations are not retroactive before the date you make your request on AIM.

Honor Code: All students are expected to adhere to the [Barnard College Honor Code](#). Approved by the student body in 1912 and updated in 2016, the Code states: *We, the students of Barnard College, resolve to uphold the honor of the College by engaging with integrity in all of our academic pursuits. We affirm that academic integrity is the honorable creation and presentation of our own work. We acknowledge that it is our responsibility to seek clarification of proper forms of collaboration and use of academic resources in all assignments or exams. We consider academic integrity to include the proper use and care for all print, electronic, or other academic resources. We will respect the rights of others to engage in pursuit of learning in order to uphold our commitment to honor. We pledge to do all that is in our power to create a spirit of honesty and honor for its own sake.*

Affordable Access to Course Texts and Materials: All students deserve to be able to study and make use of course texts and materials regardless of cost. Barnard librarians have partnered with students, faculty, and staff to find ways to increase student access to textbooks. By the first day of advance registration for each term, faculty will have provided information about required texts for each course on Canvas (including ISBN or author, title, publisher, copyright date, and price), which can be viewed by students. A number of cost-free or low-cost methods for accessing some types of courses texts are detailed on the [Barnard Library Textbook Affordability guide](#). Undergraduate students who identify as first-generation and/or low-income students may check out items from the [FLIP lending libraries in the Barnard Library](#) and in [Butler Library](#) for an entire semester. Students may also consult with their professors, the Dean of Studies, and the Financial Aid Office about additional affordable alternatives for having access to course texts. Visit the guide and talk to your professors and your librarian for more details.

Academic Accommodations: If you believe you may encounter barriers to the academic environment due to a documented disability or emerging health challenges, please feel free to contact me and/or The Center for Accessibility Resources and Disability Services ([CARDS](#)). Any student with approved

academic accommodations is encouraged to contact me via email to set up a meeting. If you have questions regarding registering a disability or receiving accommodations for the semester, please contact CARDS at (212) 854-4634, cards@barnard.edu, or learn more at the [CARDS website](#). CARDS is located in Diana Center Room 307.

Wellness Statement: It is important for undergraduates to recognize and identify the different pressures, burdens, and stressors you may be facing, whether personal, emotional, physical, financial, mental, or academic. We as a community urge you to make yourself--your own health, sanity, and wellness--your priority throughout this term and your career here. Sleep, exercise, and eating well can all be a part of a healthy regimen to cope with stress. Resources exist to support you in several sectors of your life, and we encourage you to make use of them. Should you have any questions about navigating these resources, please visit these sites:

- [Barnard Primary Care](#)
- [Barnard Counseling](#)
- [Barnard Wellness Spot](#)
- [Columbia Health](#)